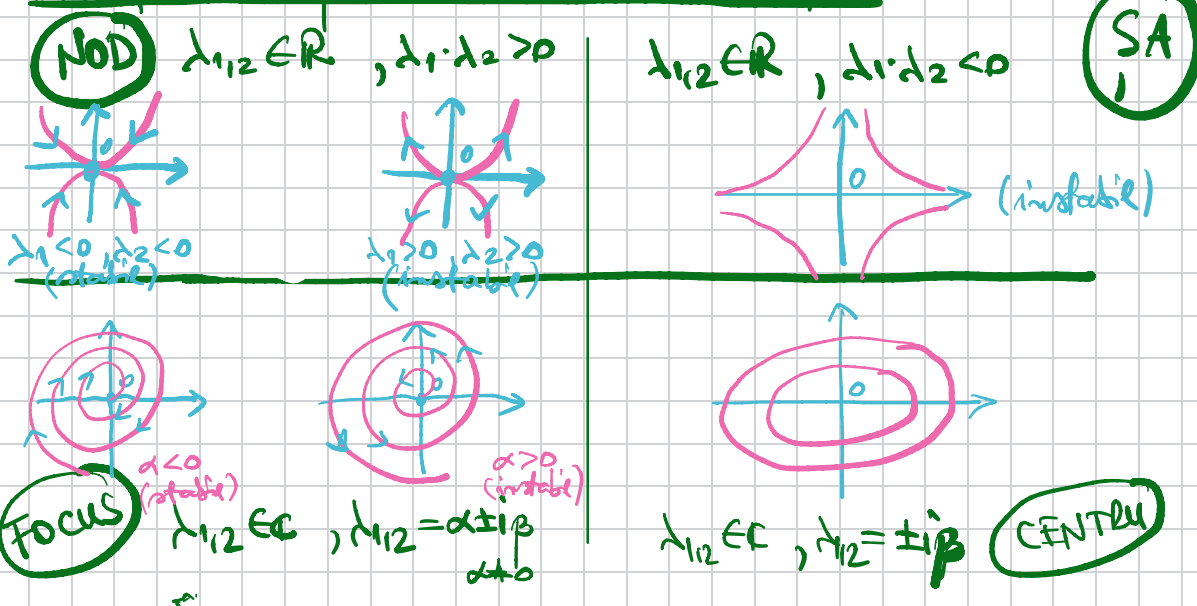


C10 - Sisteme dinamice generate de sisteme planare de ec. dif. autonome.

Exercitii

Clasificarea punctului echilibru (0,0):



Flux. P. fix. Sol. eq. Stabilitate.

①
$$\begin{cases} x' = -x \\ y' = -2y \end{cases}$$

$\Leftrightarrow \frac{dx}{dt} = -x \Leftrightarrow \frac{dx}{x} = -t \quad | \int dt$

$\Leftrightarrow \frac{dy}{dt} = -2y \Leftrightarrow \frac{dy}{y} = -2t \quad | \int dt$

$\Rightarrow \begin{cases} x(t) = c_1 \cdot e^{-t} \\ y(t) = c_2 \cdot e^{-2t} \end{cases}$

$\begin{cases} x(0) = \eta_1 \\ y(0) = \eta_2 \end{cases}$

$x(0) = c_1 \cdot e^0 = \eta_1 \Rightarrow c_1 = \eta_1$

$y(0) = c_2 \cdot e^0 = \eta_2 \Rightarrow c_2 = \eta_2$

$$\Rightarrow \begin{cases} x(t, \eta_1, \eta_2) = \eta_1 \cdot e^{-t} \\ y(t, \eta_1, \eta_2) = \eta_2 \cdot e^{-2t} \end{cases}$$

$$I_\eta = ? \quad t \in \mathbb{R} \Rightarrow I_\eta := \mathbb{R} = (-\infty, \infty); \underline{0 \in I_\eta}$$

$\hookrightarrow$ maximal

$\Rightarrow$ soluții este saturată $\Rightarrow$ definiu fluxul.

$$\varphi: I_\eta \times \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad \varphi: \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\varphi(t, \eta_1, \eta_2) = (\eta_1 \cdot e^{-t}, \eta_2 \cdot e^{-2t}) \quad \text{fluxul.}$$

Pentru a calcula orbitele facem distincții (η_1, η_2)

$$\text{I. } (\eta_1, \eta_2) = (0, 0) \Rightarrow \mathcal{S}(0_0) = \bigcup_{t \in \mathbb{R}} \varphi(t, 0, 0) = \underline{\underline{\{0, 0\}}}$$

$$\text{II } \eta_1 > 0, \eta_2 = 0$$

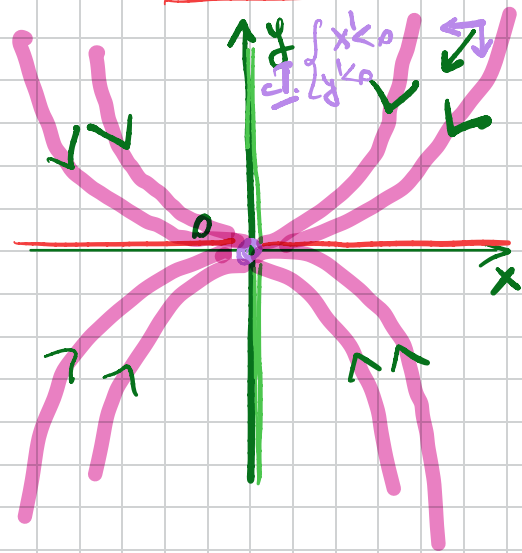
$$\begin{aligned} \Rightarrow \mathcal{S}(\eta_1, 0) &= \bigcup_{t \in \mathbb{R}} \varphi(t, \eta_1, 0) = \bigcup_{t \in \mathbb{R}} (\eta_1 \cdot e^{-t}, 0) = \\ &= \{(x, 0) \mid x > 0\} \end{aligned}$$

$$\text{III } \eta_1 < 0, \eta_2 = 0$$

$$\Rightarrow \mathcal{S}(\eta_1, 0) = \bigcup_{t \in \mathbb{R}} \varphi(t, \eta_1, 0) = \{(x, 0) \mid x < 0\}$$

$$\eta_1 \neq 0, \eta_2 = 0 \Rightarrow \mathcal{S}(\eta_1, 0) = \{ (x, 0) ; \underbrace{\left\{ \begin{array}{l} x > 0, \eta_1 > 0 \\ x < 0, \eta_1 < 0 \end{array} \right\}} \}$$

$$\begin{aligned} \text{IV. } \eta_1 &\neq 0, \eta_2 \neq 0 \\ \mathcal{S}(0, \eta_2) &= \bigcup_{t \in \mathbb{R}} \varphi(t, 0, \eta_2) = \\ &= \bigcup_{t \in \mathbb{R}} (0, \eta_2 \cdot e^{-2t}) = \\ &= \{ (0, y) ; \underbrace{\left\{ \begin{array}{l} y > 0, \eta_2 > 0 \\ y < 0, \eta_2 < 0 \end{array} \right\}} \} \end{aligned}$$



$$\begin{aligned} \text{V. } \eta_1 &\neq 0, \eta_2 \neq 0. \\ \mathcal{S}(\eta_1, \eta_2) &= \bigcup_{t \in \mathbb{R}} \varphi(t, \eta_1, \eta_2) = \bigcup_{t \in \mathbb{R}} \{ \eta_1 e^{-t}, \eta_2 e^{-2t} \} \\ &= \{ (x, y) ; y = \frac{\eta_2}{\eta_1^2} \cdot x^2 \} \end{aligned}$$

$$\text{Für } M \in \mathcal{S}(\eta_1, \eta_2) \Rightarrow \begin{cases} x_M = \eta_1 e^{-t} \\ y_M = \eta_2 e^{-2t} = \eta_2 (e^{-t})^2 = \eta_2 \cdot \frac{x_M^2}{\eta_1^2} \end{cases}$$

$$y_M = \frac{\eta_2}{\eta_1^2} \cdot x_M^2 \Rightarrow \boxed{y = k \cdot x^2} \text{ ec. orbitelor}$$

Met 2: ec. orbitelor = ?

ec. diferențiale
⇒ a orbitelor

$$\begin{cases} x' = -x \\ y' = -2y \end{cases} \Rightarrow \frac{x'}{y'} = \frac{-x}{-2y} \Rightarrow \frac{dx}{dy} = \frac{x}{2y}$$

$$2 \frac{dx}{x} = \frac{dy}{y} \Rightarrow \ln|y| = \ln|x|^2 + \ln|c|$$

$$\boxed{y = c \cdot x^2} \quad \underline{\underline{\text{ec. orbitelor}}}$$

Stabilitate:

Sist. liniar $\Rightarrow$ pct ech $(0,0) = \text{unic}$!

$$A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \quad \text{val proprii} = ?$$

$$\det |A - \lambda I_2| = 0 \Leftrightarrow \begin{vmatrix} -1-\lambda & 0 \\ 0 & -2-\lambda \end{vmatrix} = 0$$

$$\begin{cases} \lambda_1 = -1 \\ \lambda_2 = -2 \end{cases} \in \mathbb{R}, \lambda_1 \cdot \lambda_2 > 0$$

$$\begin{aligned} \lambda_1 &< 0 \\ \lambda_2 &< 0 \end{aligned}$$

$\Rightarrow (0,0) = \text{NOD}$
stabil

Temu: Analog pentru $\begin{cases} x' = x \\ y' = 2y \end{cases}$

Studiu pt. ech. stabilitate

ex2) Sistemul van der Pol:

$$\begin{cases} x' = y = f_1 \\ y' = \mu(1-x^2)y - x = f_2 \end{cases}, \underline{\underline{\mu > 0}}$$

sistem nelinier

puncte echilibru = sol sist: $\begin{cases} f_1 = 0 \\ f_2 = 0 \end{cases}$

$$\Rightarrow \begin{cases} y = 0 \\ \mu(1-x^2)y - x = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ -x = 0 \end{cases} \Rightarrow \begin{cases} y = 0 \\ x = 0 \end{cases}$$

$\Rightarrow$ Pct ech: $(x^*, y^*) = (0, 0) = \text{unic}$

sistemul linearizat: $y' = \underbrace{J_f(x^*, y^*)}_{=?} \cdot y$

$$J_f(x, y) = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1-2\mu xy & \mu(1-x^2) \end{pmatrix}$$

$$A_1 = J_f(x^*, y^*) = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}$$

val proprii ptr A_1 : $\det(A_1 - \lambda I_2) = 0$.

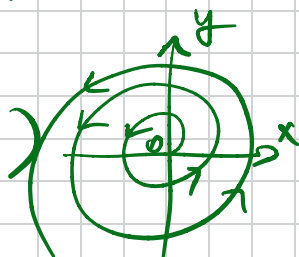
$$\begin{vmatrix} -\lambda & 1 \\ -1 & \mu - \lambda \end{vmatrix} = 0 \Rightarrow \lambda_{1,2} = \frac{\mu \pm \sqrt{\mu^2 - 4}}{2}$$

$(\mu > 2)$

$0 < \mu < 2 \Rightarrow \lambda_{1,2} \in \mathbb{C} \setminus \mathbb{R}$

$$\operatorname{Re} \lambda_{1,2} = \frac{\mu}{2} > 0$$

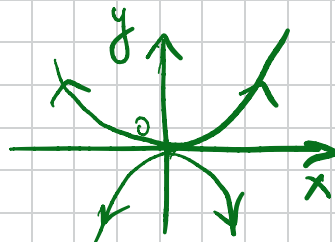
$\Rightarrow (0,0) = \text{focar instabil}$
(spirală instabilă)



$\mu > 2 \Rightarrow \lambda_{1,2} \in \mathbb{R}_+$

$$\lambda_1 > 0, \lambda_2 > 0, \lambda_1 \cdot \lambda_2 > 0$$

$\Rightarrow (0,0) = \text{nod instabil}$



Tema: Pendulul fără frecare: $\begin{cases} x' = y \\ y' = -\sin x \end{cases}$

indicatii $\rightarrow$ discută $x = (2k)\pi, x = (2k+1)\pi$ la Jacobian

Tema:
$$\begin{cases} x' = y + x - x^3 \\ y' = -x + y - y^3 \end{cases}$$

indicatii rezolvare
$$\begin{cases} f_1 = 0 \\ f_2 = 0 \end{cases}$$

⊕

↓

da factor comun

(0,0)

(±1,0)

(0, ±1)

Tema:
$$\begin{cases} x' = x - x^3 \\ y' = -y \end{cases}$$

(0,0)

(±1,0)

21.3
$$\begin{cases} x' = y + x(x^2 + y^2 - 1) \\ y' = -x - y(x^2 + y^2 - 1) \end{cases}$$

pt. eq.:
$$\begin{cases} f_1 = 0 \\ f_2 = 0 \end{cases} \Leftrightarrow \begin{cases} y + x(x^2 + y^2 + 1) = 0 \\ -x - y(x^2 + y^2 + 1) = 0 \end{cases} \textcircled{+}$$

$$y + x(x^2 + y^2 + 1) - x - y(x^2 + y^2 + 1) = 0.$$

$$(y - x) - (x^2 + y^2 + 1)(y - x) = 0.$$

$$(y - x)(-x^2 - y^2) = 0 \Leftrightarrow (y - x)(x^2 + y^2) = 0$$

$$\Rightarrow y = x \quad \leadsto \text{eq (1): } y + y(y^2 + y^2 + 1) = 0.$$

$$y(2y^2 + 2) = 0 \Rightarrow \begin{matrix} y^* = 0 \\ x^* = 0 \end{matrix}$$

$$\Rightarrow (x^*, y^*) = (0, 0)$$

$$J(x, y) = \begin{pmatrix} +3x^2 + y^2 + 1 & 1 + 2xy \\ -1 - 2xy & -x^2 - 3y^2 - 1 \end{pmatrix}$$

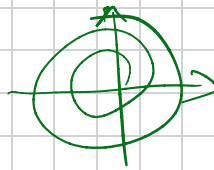
$$J(0,0) = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = A_1$$

$$\det(A_1 - \lambda I_2) = 0.$$

$$\begin{vmatrix} 1-\lambda & 1 \\ -1 & -1-\lambda \end{vmatrix} = 0.$$

$$-(1-\lambda)(1+\lambda) + 1 = 0$$

$$\lambda^2 = 0 \Rightarrow \lambda_{1,2} = 0$$



$$\operatorname{Re} \lambda_{1,2} = 0.$$

stabilitatea
neconcluzivă

T. primă
aprox.

Modelul pradă-prădător (Lotka - Volterra)

(Cap. 10)

$x = x(t)$ = pop. pradă ($t \geq 0$)

$x_0 = x(t_0)$

$y = y(t)$ = pop. prădător ($t \geq 0$)

$y_0 = y(t_0)$

$t_0 = 0$ momentul inițial

Ipoteze:

- ① hrana pradă = nelimitată } $\Rightarrow$ Malthus
prădătorul lipsesc } ptr pradă

$$x'(t) = a \cdot x(t) \quad , \quad a > 0$$

- ② prădătorul nu are alte resurse hrănă
dacă pradă lipsesc $\Rightarrow$ Malthus ptr
prădător

$$y'(t) = -c y(t) \quad , \quad c > 0$$

③ dacă $\exists$ pradă $\Rightarrow$ pradă este influențată de nr. de interacțiuni pradă-predator

Nr. interacțiuni: $x(t) \cdot y(t)$

$\Rightarrow$ rata de creștere a pp. pradă:

$$x'(t) = a \cdot x(t) - b \cdot x(t) y(t), a, b > 0$$

④ dacă $\exists$ pradă $\Rightarrow$ pradătorul este influențat de nr. de interacțiuni p/p
 $\Rightarrow$ rata de creștere a pp. pradător:

$$y'(t) = -c y(t) + d x(t) \cdot y(t), c, d > 0$$

$$\Rightarrow \begin{cases} x'(t) = a x(t) - b x(t) y(t) \\ y'(t) = -c y(t) + d x(t) y(t) \end{cases}$$

$$a, b, c, d > 0$$